

Hongrui (Jack) Zhang

Computer Science Ph.D. Researcher | Data Systems and GPU Computing

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EDUCATION

Ph.D. in Computer Science, University of California, Riverside

Sep 2024 – Expected Jun 2028

Advisor: Hung-Wei Tseng

B.S. in Computer Engineering, University of California, San Diego

Sep 2019 – Sep 2023

RESEARCH EXPERIENCE

GPU-Accelerated Database Systems and Matrix-Native Query Processing

Jun 2026 – Present

Python, PyTorch, SQL, CUDA, cuSPARSE | Query execution and experimental evaluation

- Developed and validated a matrix-native execution plan for TPC-H Q3, expressing joins, grouping, and aggregation using sparse matrix and tensor operations.
- Optimized TCUDB-style sparse joins using direct one-hot CSR construction, reducing Q4 join latency from 40.5 ms to 11.0 ms ($3.7\times$) over generic COO construction at TPC-H SF10.
- Implemented and SQL-validated PyTorch execution plans for all 22 TPC-H queries, profiling GPU bottlenecks on an RTX 5090 to guide hardware-aware query processing.

RTSpMSpM — RT-Core-Accelerated Sparse Matrix Multiplication

Sep 2023 – Sep 2025

ISCA '25 | First author and conference presenter | OptiX, CUDA, C++, Python

- Designed and implemented an OptiX/CUDA library mapping sparse matrix multiplication onto NVIDIA RT Cores, achieving a $1.85\times$ geometric-mean over cuSPARSE.
- Proposed RT core architectural enhancements, including fused intersection-multiplication and hardware row accumulation, achieving $3.06\times$ simulated speedup over cuSPARSE with only 0.2% estimated area overhead.
- Integrated the library into a graph neural network training framework using CUDA and Python, achieving more than $3\times$ speedup on NVIDIA RTX 4090.

PUBLICATION

Hongrui Zhang, Yunan Zhang, and Hung-Wei Tseng. "RTSpMSpM: Harnessing Ray Tracing for Efficient Sparse Matrix Computations." *52nd International Symposium on Computer Architecture (ISCA '25)*, 2025. [DOI]

IEEE Micro Top Picks 2026 — **Honorable Mention**

TECHNICAL SKILLS

Languages: Python, C++, CUDA C++, SQL, C, Bash

Frameworks & Tools: PyTorch, CUDA, NVIDIA OptiX, cuSPARSE, NumPy, Git, Docker, Linux

GPU Profiling & Data Systems: Nsight Systems, Nsight Compute, PyTorch Profiler, TPC-H, sparse matrix operations, query execution